



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 • STANDARD 6.G.1

Area of Composite Figures

Lesson 5-5 • Teacher Copy

Name: Type your name

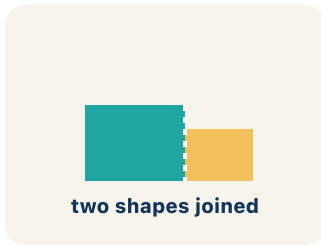
Date: Today's date

Class: Class period

Key Vocabulary

Level 2 Standard

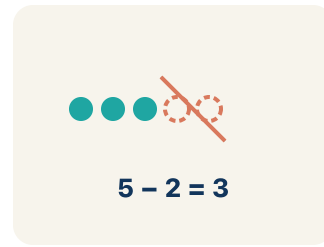
Picture first, then the word, then a plain-language meaning. Say each word out loud.



An L-shaped room = a 12×8 rectangle joined to a 6×5 rectangle; total area = $96 + 30 = 126$ sq ft

Composite Figure

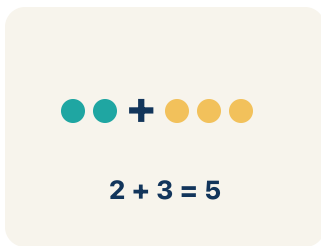
Write the definition:



Draw a dashed line across the L-shape to split it into two rectangles — now you can find each area separately

Decompose

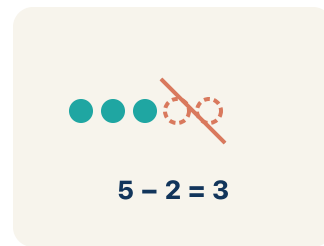
Write the definition:



T-shaped hallway: top rectangle = 30 sq ft, bottom rectangle = 28 sq ft → total = $30 + 28 = 58$ sq ft

Add

Write the definition:



A 14×10 pool with a 6×4 cutout: $140 - 24 = 116$ sq ft of water surface

Subtract

Write the definition:

$$3x + 5$$

no equal sign

Rectangle: $A = l \times w$; Triangle: $A = \frac{1}{2} \times b \times h$; use the right formula for each piece of a composite figure

Formula

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the area of a composite figure by adding or subtracting the areas of basic shapes.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Composite Figure

Decompose

Add

Subtract

Formula



Tap any word to see what it means and a picture.

1

A shape made of two or more basic shapes combined is a

2

To break a composite figure into basic shapes is to it.

3

To find the total area of separate parts, I their areas.

4

To find the area of a shape with a piece removed, I the missing part's area.

5

A math rule written with symbols is a .



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: An L-shaped room is made of two rectangles: $10\text{ ft} \times 6\text{ ft}$ and $4\text{ ft} \times 3\text{ ft}$. What is the total area?

- A. 72 sq ft
- B. 60 sq ft
- C. 12 sq ft
- D. 72 ft

Step 1 $\text{Area } 1 = 10 \times 6 = 60\text{ sq ft.}$

Step 2 $\text{Area } 2 = 4 \times 3 = 12\text{ sq ft.}$

Step 3 $\text{Total} = 60 + 12 = 72\text{ sq ft.}$

 **Answer:** A. 72 sq ft

 Try — put the steps in order

Drag the cards (or use the ▲ ▼ buttons) to put the solution steps in the right order, then press **Check**.

Total = $60 + 12 = 72\text{ sq ft.}$

Area 1 = $10 \times 6 = 60\text{ sq ft.}$

Area 2 = $4 \times 3 = 12\text{ sq ft.}$

 We do — together

Solve this one with your class using the same steps.

Problem: A rectangular patio is $15\text{ ft} \times 10\text{ ft}$ with a $5\text{ ft} \times 4\text{ ft}$ rectangular flower bed cut out. What is the remaining area?

- A. 130 sq ft
- B. 150 sq ft
- C. 20 sq ft
- D. 170 sq ft

Step 1

Step 2

Step 3

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

- A. 23 sq units
- B. 15 sq units
- C. 8 sq units
- D. 20 sq units

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Plan 2 — L-shaped lobby: decompose the L into two rectangles, one 9 ft by 6 ft and one 7 ft by 3 ft, then ADD. What is the total floor area?

- A. 75 sq ft
- B. 63 sq ft
- C. 54 sq ft
- D. 21 sq ft

Show your work:

2. Plan 4 — Design studio: the floor is a 16 ft by 12 ft rectangle, but a 5 ft by 4 ft storage closet is cut out of one corner. Find the usable floor area.

- A. 172 sq ft
- B. 192 sq ft
- C. 212 sq ft
- D. 20 sq ft

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A T-shaped room can be decomposed in two different ways. Show both ways and prove they give the same total area. Use a T-shape where the top is $14\text{ ft} \times 4\text{ ft}$ and the stem is $6\text{ ft} \times 10\text{ ft}$.

Sentence starter: Way 1: I split it into ____ and ____ . Areas: ____ + ____ = ____ . Way 2: I used a ____ rectangle and subtracted ____ . Areas: ____ - ____ = ____ . Both give ____ sq ft.

Show your work:

Reflect — Exit Ticket

A composite figure is made of a $9\text{ ft} \times 7\text{ ft}$ rectangle and a $3\text{ ft} \times 4\text{ ft}$ rectangle joined together. What is the total area?

- A. 75 sq ft
- B. 63 sq ft
- C. 12 sq ft
- D. 75 ft

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Composite Figure
2. **Notes 2:** Decompose
3. **Notes 3:** Add
4. **Notes 4:** Subtract
5. **Notes 5:** Formula
6. **We Try (notes):** A. 130 sq ft — *Patio = $15 \times 10 = 150$ sq ft. Cutout = $5 \times 4 = 20$ sq ft. Remaining = $150 - 20 = 130$ sq ft.*
7. **You Try (notes):** A. 23 sq units — *$5 \times 3 = 15$. $2 \times 4 = 8$. Total = $15 + 8 = 23$ sq units.*
8. **Try It 1:** A. 75 sq ft — *First rectangle = $9 \times 6 = 54$ sq ft. Second rectangle = $7 \times 3 = 21$ sq ft. Joined L-shape, so add: $54 + 21 = 75$ sq ft.*
9. **Try It 2:** A. 172 sq ft — *Large rectangle = $16 \times 12 = 192$ sq ft. Cutout closet = $5 \times 4 = 20$ sq ft. A piece is removed, so subtract: $192 - 20 = 172$ sq ft.*
10. **Exit Ticket:** A. 75 sq ft — *Area 1 = $9 \times 7 = 63$ sq ft. Area 2 = $3 \times 4 = 12$ sq ft. Total = $63 + 12 = 75$ square feet.*